GE HEALTHCARE

Technical Publications

GE Part Number 5140621

Revision 14

TEAL PDU Module in XGD Cabinet

Operating Documentation

TABLE OF CONTENTS

IMPORTANT SAFETY INSTRUCTIONS

SAVE THESE INSTRUCTIONS – This manual contains important instruction for the XGD Power Distribution Unit (PDU) that must be followed during installation, operation, and maintenance.

DANGER

Electrical shock hazard. Opening enclosure exposes hazardous voltages. Refer service to qualified personnel.

IMPORTANT

As standards, specifications, and designs change from time to time, please ask for confirmation of the information given in this publication.

1- INTRODUCTION

1.1 General Description

The XGD Power Distribution Unit (PDU) distributes power to various components of a General Electric Medical Systems MR system while providing surge suppression and electrical isolation for the power line. Three phase input voltage is selectable among 380/400/415/480 VAC at a rated frequency of 50/60 Hz. The power rating for this PDU is 79 kVA continuous and 103 kVA instantaneous. The PDU mounts into the GE 5267500 cabinet.

XGD Power Distribution Unit
Illustration 1-1

1-2 Scope

This manual describes procedures for installation and basic operation of the XGD Power Distribution Unit as well as procedures for failure diagnosis/isolation and installation of field replaceable components of the PDU.

2- SPECIFICATIONS

GE Specifications: 5140621PSP

<u>GE Part Number</u>
5140621

2.1 Electrical Specifications

Power Rating

79 kVA Continuous; 103 kVA Instantaneous

Input Voltage

380/400/415/480 VAC Delta.

Output Voltage

208/120 VAC WYE and 242/420 VAC Wye

Output Circuit Breaker and Connection Requirements

Breaker Size	**Function**	**Ref. Desig**	**Sec Voltage**	**Wire to Phase**	**E-Stop**	**Customer Access**	**Connection Type**
90A 3 Pole	Gradient 420V Power	CB2	420	ABC	Yes	Front	APP#1321G4, G5(3 Connectors)
70A 3 Pole	PDM – RF	CB3	208	ABC	Yes	Rt. Side	Breaker
25A 3 Pole	Spare	CB4	208	ABC	No	Top Left Side	Breaker
15A 1 Pole	Brainwave	CB5	120	A	No	Top Left Side	Breaker
20A 1 Pole	Host	CB6	120	A	No	Top Left Side	Breaker
32A 3 Pole	Supercon Shim PS	CB7	208	ABC	No	Front	5 wire twist lock (Nema style L21-30R Hubbell) (receptacle front of cab)
32A 3 Pole	Magnet Power Supply	CB8	208	ABC	No	Front	5 wire twist lock (Nema style L21-30R Hubbell) (receptacle front of cab)
10A 2 Pole	Gradient 48V Control	CB9	208	BC	No	Top Rt.	7W2 D-Sub connectors
20A 2 Pole	PDM – Recon	CB10	208	BC	No	Rt. Side	Breaker

Overload Protection

Input main circuit breaker (CB1) with adjustable trip settings. Output distribution circuit breakers.

Load Current

83 Amps total maximum continuous (208V)
67 Amps total maximum continuous (420V), 103A Peak 5 sec Max.

Transformer Impedance

Less than 2%

Efficiency

98% Typical

2.2 Environmental Specifications

Heat Dissipation

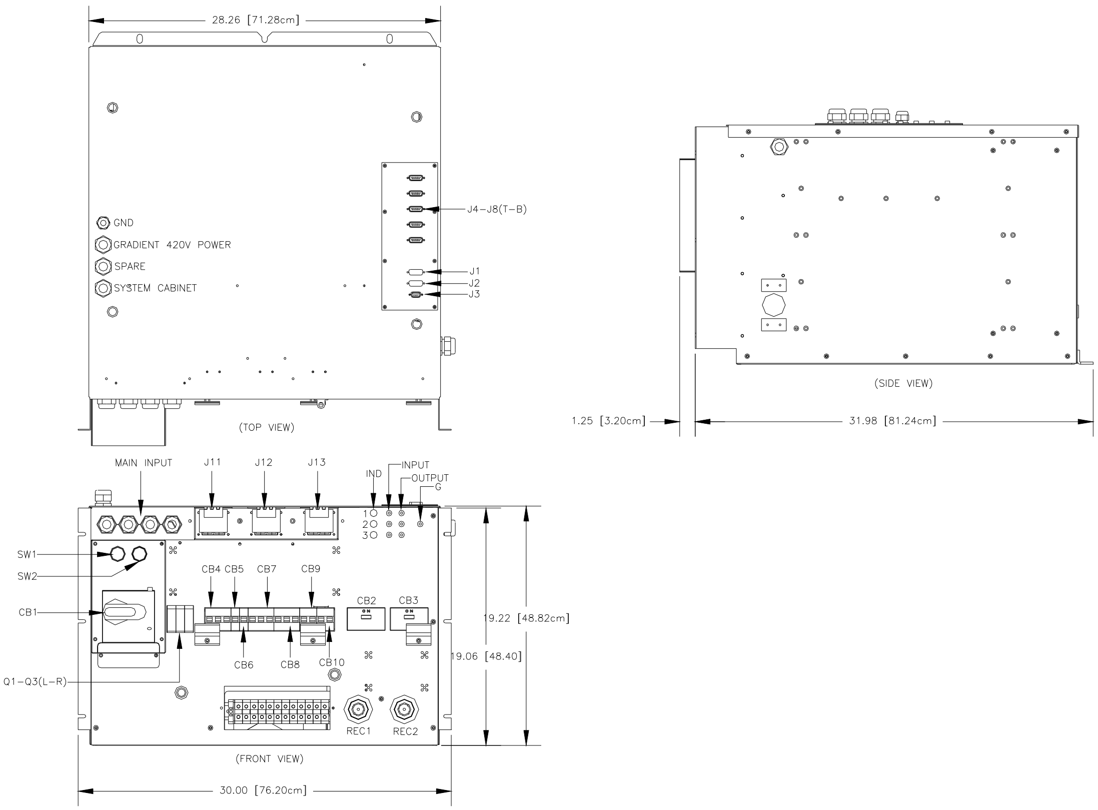
4,076 BTU/Hr

2.3 Mechanical Specifications

Weight

879 lbs. / 398 kg.

Dimensions



XGD Overall Dimensions
Illustration 2-1

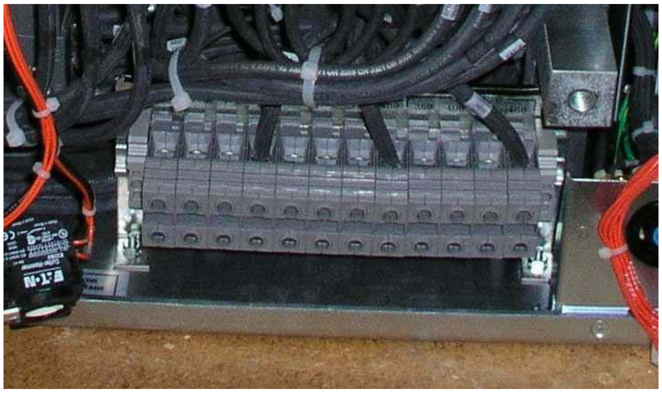
3- INSTALLATION AND OPERATION

3.1 Installation

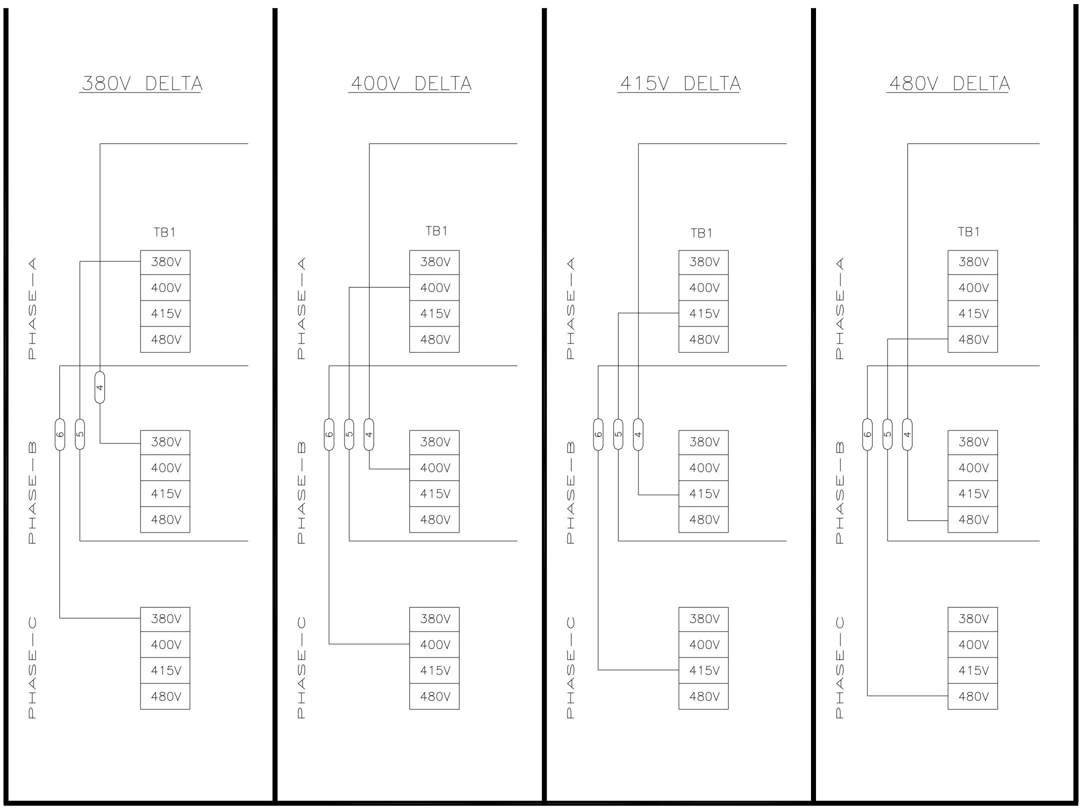
3.1.1 Input Voltage Selection

Before connecting the input power cables, determine the nominal value of the input voltage. The PDU allows for nominal input voltages of 380, 400, 415, or 480 VAC (three phase). Input voltage selection is accomplished as follows:

1. On the Main Disconnect Panel (MDP), turn off the PDU circuit breaker, lock and tag appropriately.
2. Using a voltmeter or other voltage indicating device at the Main Input Terminal Board, check to be sure that no voltage is being applied to the input of the PDU.
3. Remove the small cover panels from the front panel of the PDU marked “INPUT VOLTAGE SELECT”.
4. Illustration 3-1 shows the Input Voltage Selection terminal block (TB1). For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.



INPUT VOLTAGE SELECT TERMINAL BLOCK (TB1)
Illustration 3-1



INPUT VOLTAGE SELECTION
Illustration 3-2

5. The overload and short circuit trip settings for the Input Circuit Breaker (CB1) must be set to the correct values corresponding to the input voltage. See Table 3-1.

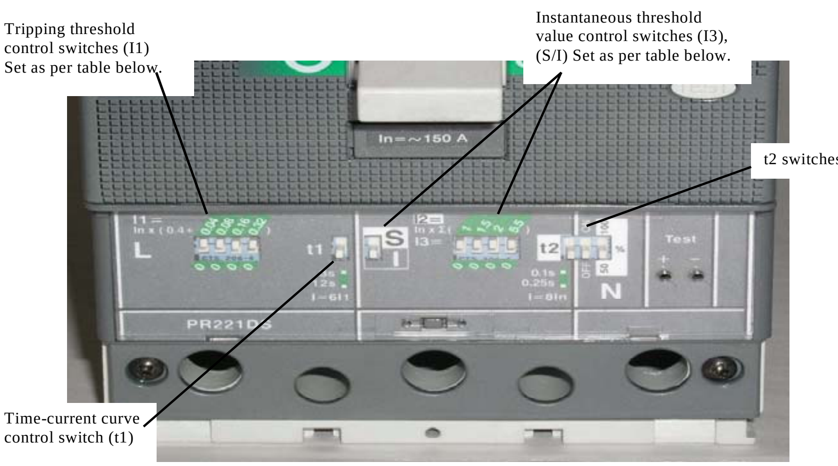
5a. Remove the small frame covering the main circuit breaker and the small panel underneath marked “CURRENT ADJUST”. Open the transparent plastic cover on the lower portion of the circuit breaker to give access to the dip switches controlling the circuit breaker operation.

5b. Illustration 3-3 shows the position of the dip switches for each input voltage selection. The “L” and “I” switches must match the configuration for the incoming voltage used as shown in illustration 3-2.

Maximum Input Current vs. Input Voltage

Table 3-1

Input Voltage	380V	400V	415V	480V
Maximum Input Current	121A	115A	111A	96A



SWITCH SETTINGS FOR CB1

INPUT VOLTAGE	380	400	415	480
CB AMP SIZE	150A	150A	140A	120A
CB ”L” SETTINGS	0.60	0.60	0.56	0.48
”L” SWITCH SETTINGS				
”t1” & ”S/I” SETTINGS				
”I3” SWITCH SETTINGS				
”t2” SWITCH SETTINGS				

CIRCUIT BREAKER DIP SWITCH SETTINGS
Illustration 3-3

3.1.2 Connection

Note the Cable Access Panel shown in Illustration 3-5 above the main circuit breaker. This panel provides strain relief cable clamps for the input and the various output cables. This panel may be removed from the PDU to provide easier access to the terminal blocks.

1. Route the cables through their respective cable clamps and connect cables to the appropriate terminals as shown.
2. Replace the Cable Access Panel and tighten the cable clamps.

IMPORTANT

For input cables, use wire sizes in accordance with National Electric Code Table 310-16. The following table gives maximum full-load input currents in amperes (which would occur at 10% undervoltage) for each of the input voltages, as a guide to selecting proper wire sizes.

3.2 Operation

3.2.1 Main Input Circuit Breaker

The Main Input Circuit Breaker operating handle has three positions:

- OFF (as shown in Illustration 3-4),
- ON (vertical), and
- TRIPPED (slightly to the right of vertical). If the Main Input Circuit Breaker has been tripped, it is necessary to move the handle to the OFF position to reset it before turning it ON.

MAIN INPUT CIRCUIT BREAKER OPERATING HANDLE
Illustration 3-4

The Main Input Circuit Breaker may be locked in the OFF position. By pressing on the small triangle on the operating handle, the lockout tab is made accessible to a padlock.

3.2.2 Power Off

This pushbutton at the top of the front panel, when pressed, will immediately trip the Main Input Circuit Breaker, interrupting all power to the load.

3.2.3 EMO Reset

This pushbutton immediately to the right of the Power Off pushbutton will enable the Emergency Stop circuit.

3.2.4 TVSS Indicators

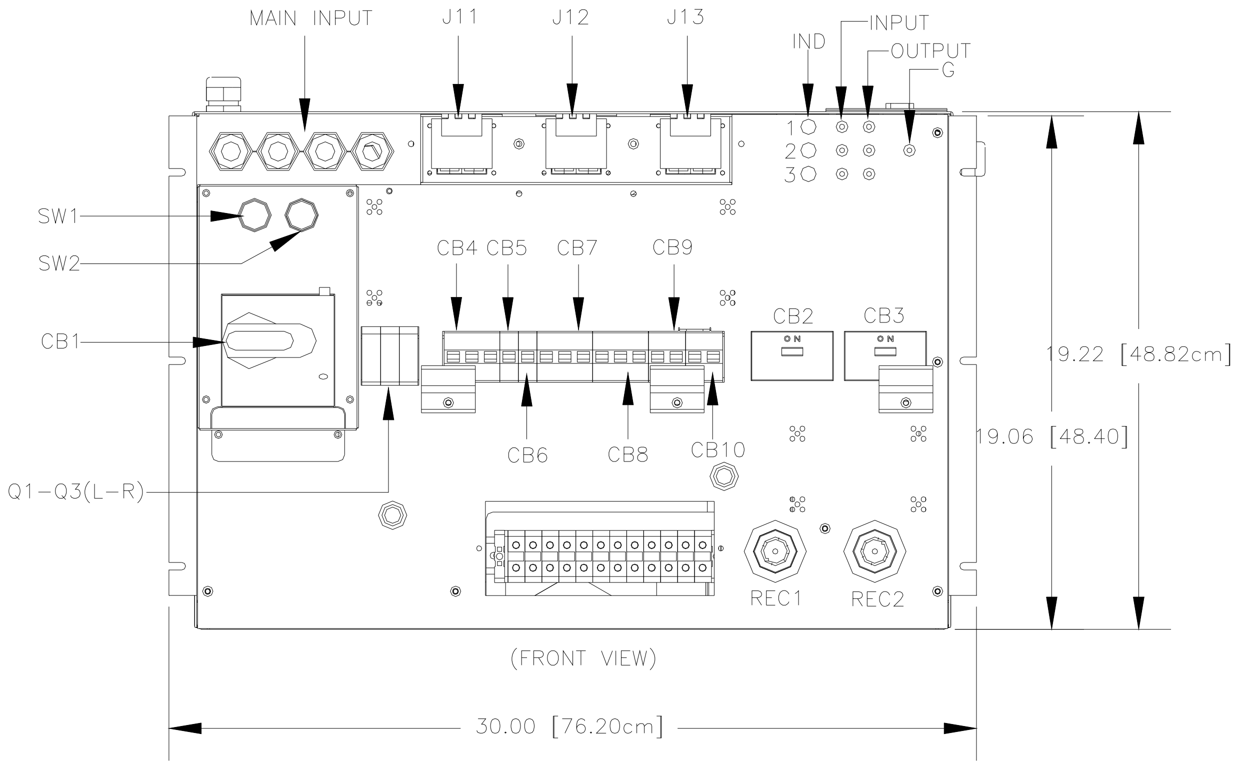
To the right of the Main Input Circuit Breaker are three TVSS (Transient Voltage Spike Suppressor). Any module that fails must be replaced to restore the PDU to operation with surge protection functioning. See Section 4-1.1, Replacement of a TVSS Module, for instructions for replacing a TVSS Module.

3.2.5 Phase Indicators

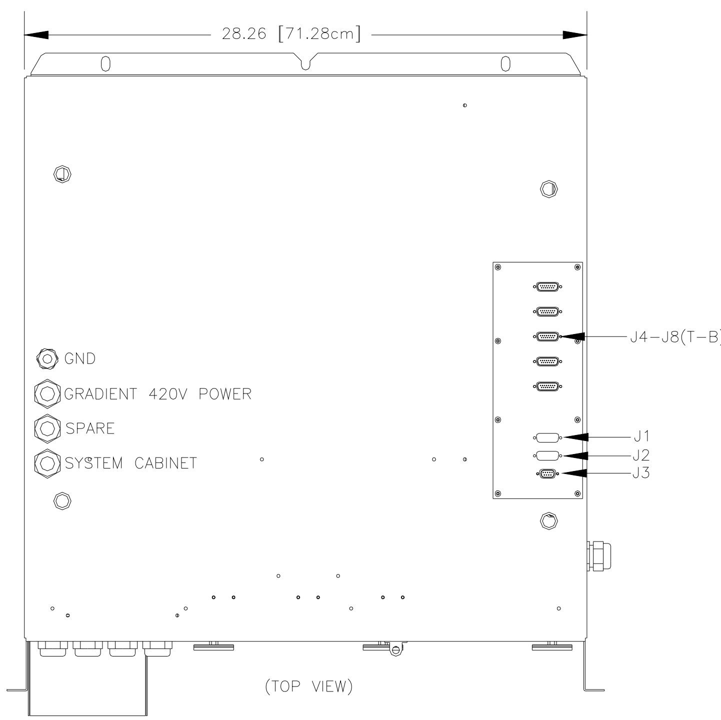
There are three Phase Indicator lights. These indicate that the transformer is energized and ready to deliver power to the load.

3.2.6 Input / Output Voltage Test Points

Near the Phase Indicators on the front panel are jacks into which can be plugged connectors for testing for input and output voltage. See Illustration 3-5. The voltage at these points are reduced by a factor of 100:1 from the actual voltage at the terminal blocks.



FRONT PANEL COMPONENT LOCATIONS
Illustration 3-5



TOP PANEL COMPONENT LOCATIONS
Illustration 3-6

4- FIELD REPLACEABLE UNITS (FRUs)

LISTING OF FRUs
4-1

Item	Ref Desig	Description	GE P/N	TEAL P/N
1	A1	Control PC Board Asm	5140621-2	4990780
2	Q1-Q3	TVSS Module	2301200-3	4990387
3	CON1-CON2	420V E-Stop Contactor	5140621-3	4990781
4	PS1-PS2	48V Power Supplies	5140621-4	4990782
5	FAN1-FAN2	48V Fan with Tach	5140621-5	4990783

4.1 FRU Installation

4.1.1 Replacement of the TVSS Module

The operation of the TVSS Indicators is explained in Section 3-2.4, TVSS Indicators. If one (or more) of the indicators shows that a TVSS Module has failed, follow the steps below to replace the failed module and restore TVSS protection:

1. Turn off the PDU circuit breaker, lock and tag.
2. Using a voltmeter or other voltage indicating device, check the voltage at the Main Input Terminal Board to be sure that no voltage is being applied to the input of the PDU.
3. Pull out the failed TVSS Module(s) and plug in the replacement(s) as shown in Illustration 4-1.

TVSS MODULE REPLACEMENT
Illustration 4-1

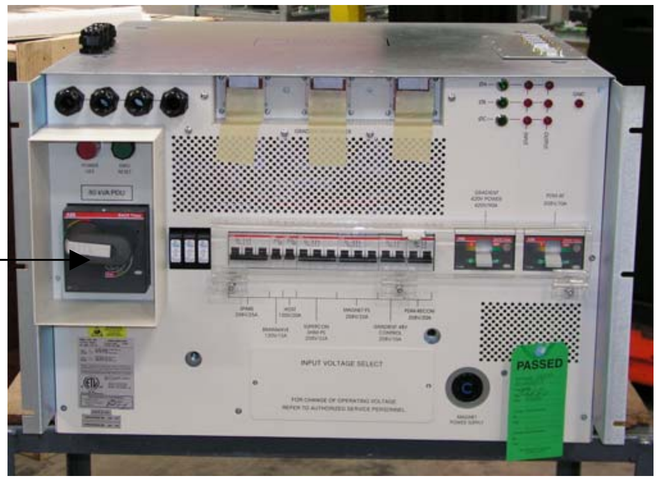
4. Re-energize the PDU.
5. Verify that all TVSS Modules are good (no red flags displayed).

4.1.2 Removal of Front Panel to Access Additional FRUs

Tools needed: Phillips screwdriver or power screwdriver set, 11/32 open wrench.

1. Turn off the main input circuit breaker, lock and tag.

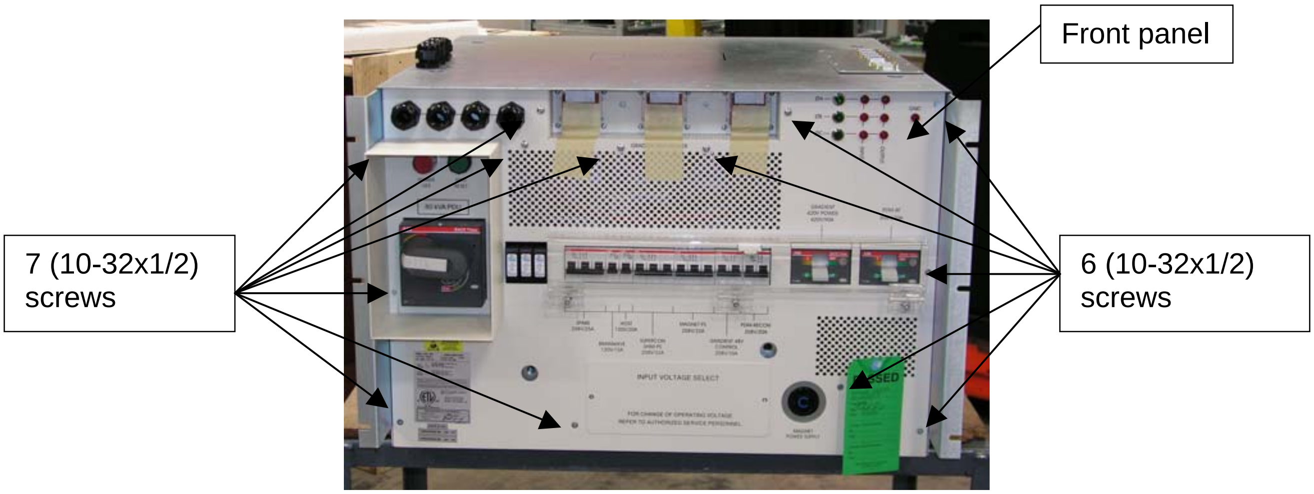
Main Input Circuit Breaker



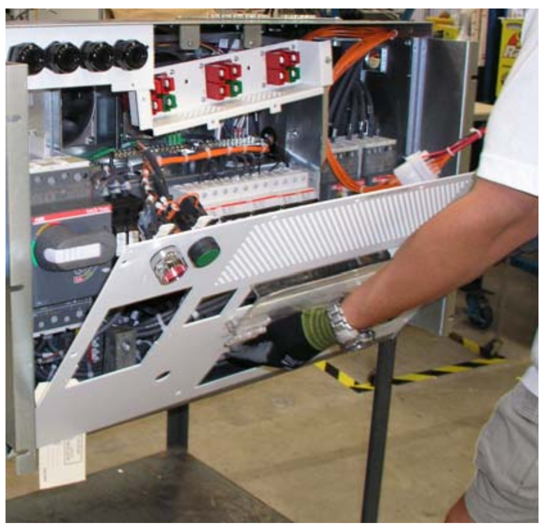
2. Remove 4 (6-32 x 3/8) screws on the circuit breaker guard.
   Note: Do not discard screws, as they will be used to put back guard.

4 (6-32 x 3/8) screws

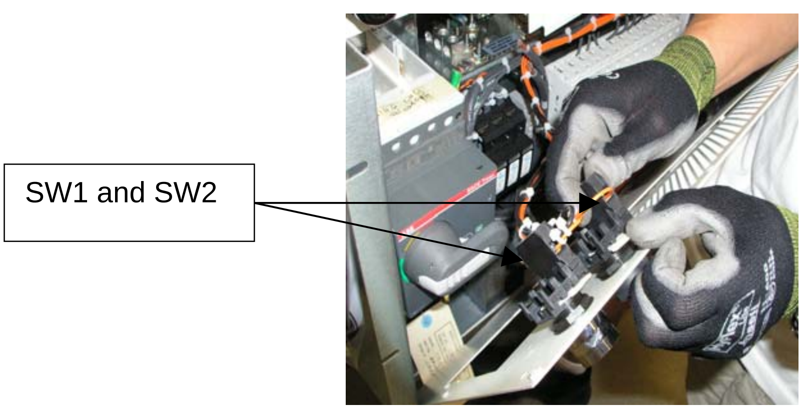
3. Remove 13 (10-32x1/2) screws on the front panel using Phillips screwdriver.
   Note: Do not discard screws, as they will be used to put back front plate.



4. Slightly pull the front panel and lean it toward the front.



5. Disconnect SW1 and SW2 from the back of the front panel. Note: No tools will be required, turn tab counterclockwise and gently pull assembly apart.



6. Disconnect P8 and J8 15Pin housing connector. Note: No tools required, press locking tab and pull connectors apart.

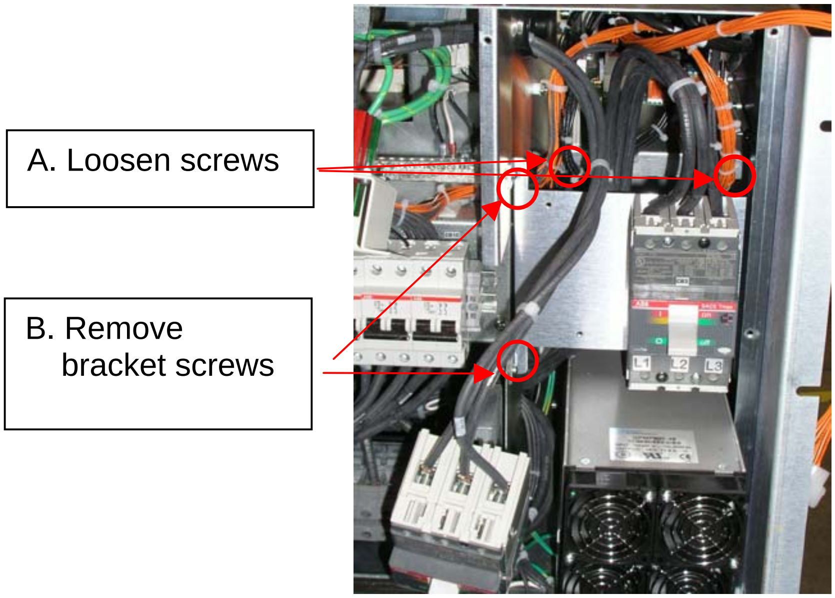
4.1.3 Replacement of the Control PC Board

If the control board is deemed defective as per the Troubleshooting Guide in 4-1.8 follow the steps below to replace the control board. The control board is mounted on the inside of the side panel.

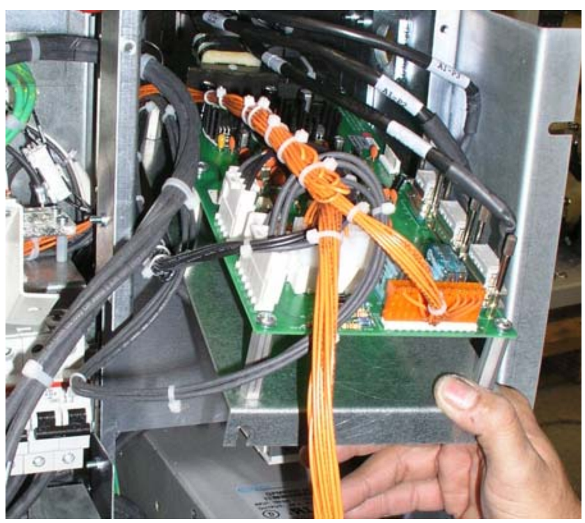
Circuit boards or assemblies with static sensitive devices will be packaged in accordance to prevent damage due to ESD (Electrostatic Discharge). Procedures to prevent ESD should be followed when handling static sensitive devices. Wear protective gloves for sharp edges.

1. Remove the four mounting screws from the circuit breakers (CB1 and CB2).
   Note: Do not discard screws, as they will be used to replace the CBs.

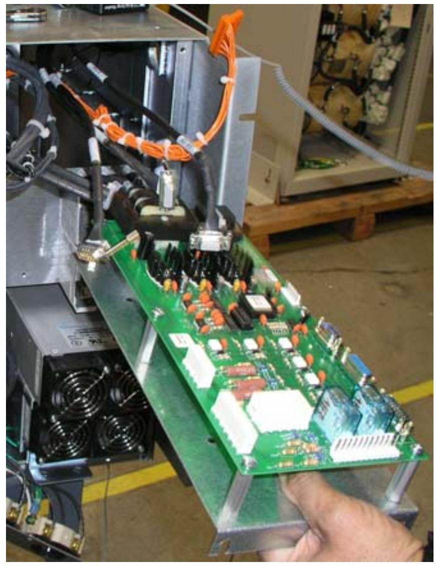
2. Loosen (do not remove) the 2 screws (10-32x1/2) at PCB (see A in photo below).



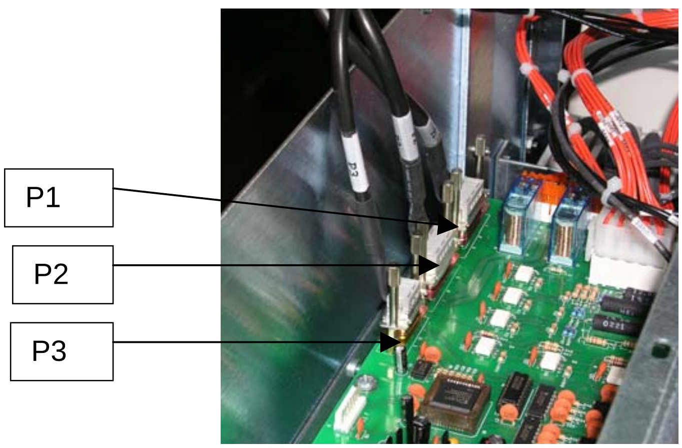
3. Remove left two screws from bracket (see B in photo above). Cut tie wraps as needed to remove PCB. Wear protective gloves for sharp edges.
4. Carefully lift up PCB assembly and pull out half way.

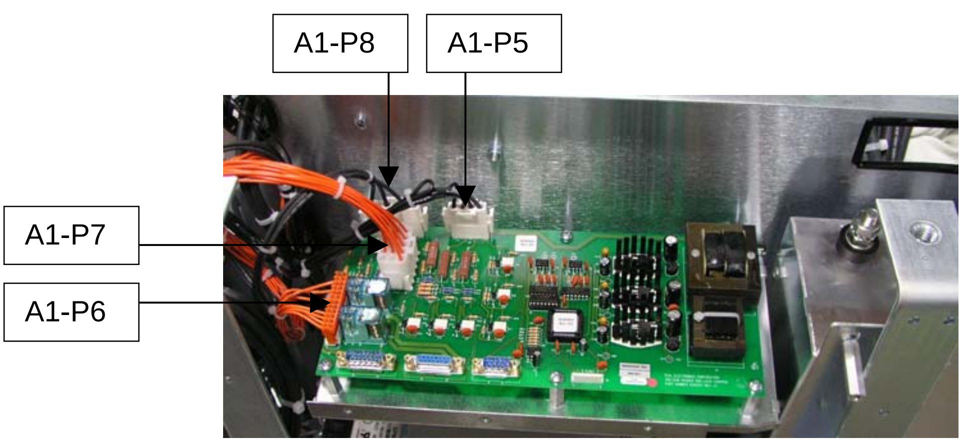


5. Disconnect cables P1, P2, P3, A1-P5, A1-P6, A1-P7, A1-P8. Mark the cables with a masking tape and marker to ensure correct position when re-connecting the cables.



The following two photos are to be used for reference of placement of each cable.





6. Insert and connect the new board.

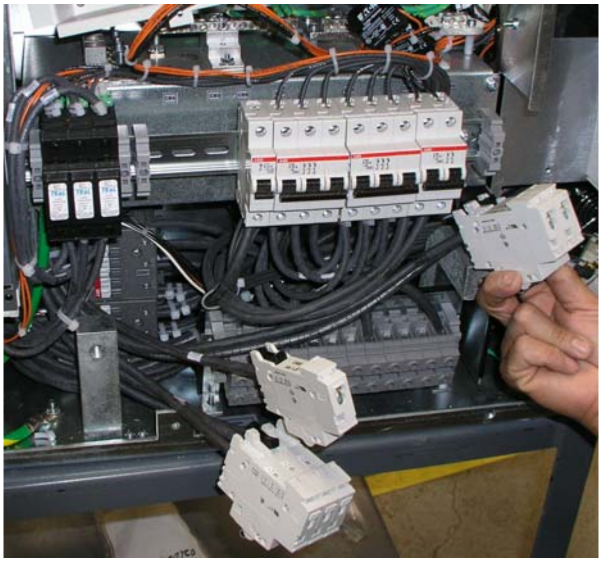
4.1.4 Replacement of Contactors (CON1 & CON2)

If the contactors are deemed defective as per the Troubleshooting Guide in 4-1.8 follow the steps below to replace the contactors. The contactors are mounted behind the circuit breaker panel of the PDU located on the front panel.

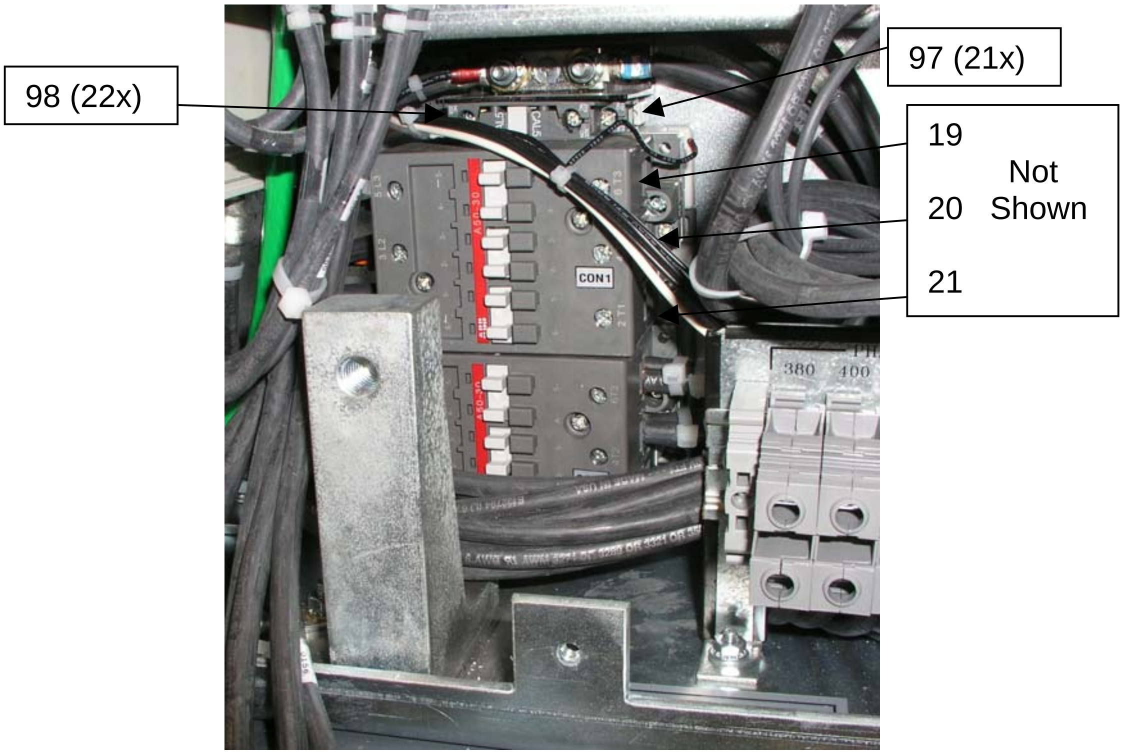
Contactors or assemblies with static sensitive devices will be packaged in accordance to prevent damage due to ESD (Electrostatic Discharge). Procedures to prevent ESD should be followed when handling static sensitive devices.

Tools needed: Phillips screwdriver or power screwdriver set, slotted screwdriver, 11/32 open/closed wrench.

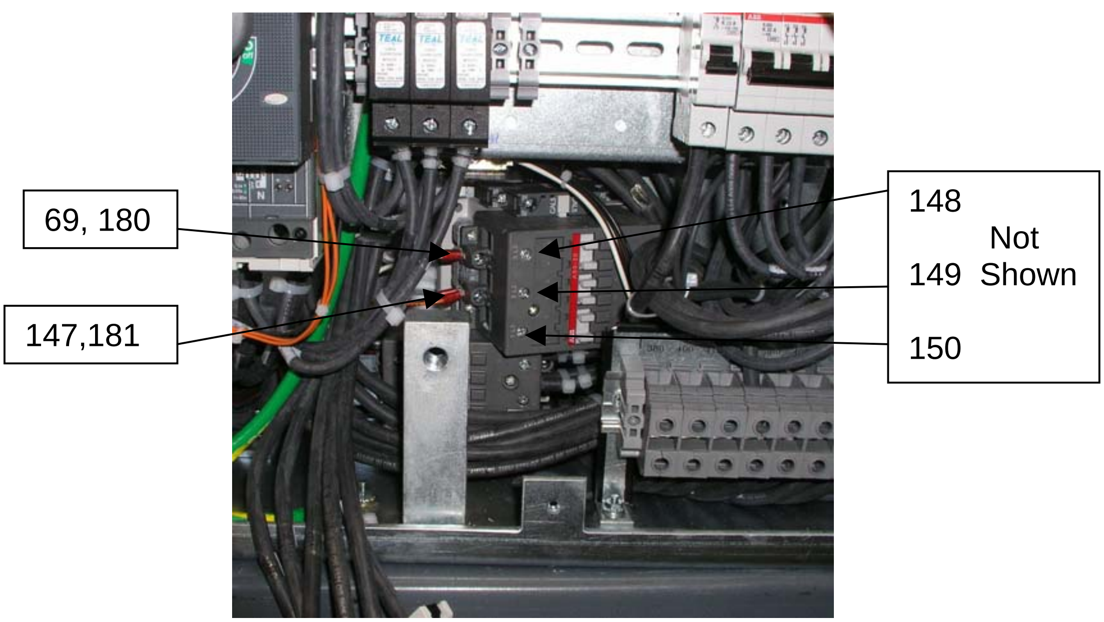
1. Remove (snap off) circuit breakers (CB4, CB5 and CB10) by twisting breaker upwards from the mounting bracket.



2. Disconnect wires 97, 98 at the AUX of CON1 and wires 19 – 21 at T1-T3 of CON1. Make sure you mark the wires with a masking tape and marker.

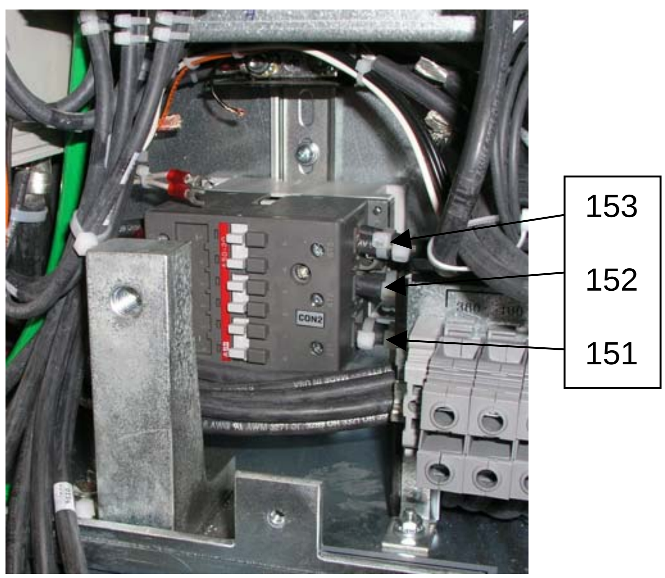


3. Snap out CON1 from the mounting bracket and disconnect wire 148 – 150 at L1 – L3.
4. Disconnect wires 69/180 at A1 of CON1 and wire 147/181 at A2 of CON1. Make sure you mark the wires with a masking tape and marker.



5. Remove CON1 from unit and proceed to removing CON2.

6. Disconnect wire 151 at L1, 152 at L2 and 153 at L3 of CON2.



7. Snap out CON2 and disconnect wires 128/63 at T1, 65/129 at T2 and 130/66 at T3 of CON2.
8. Disconnect wire 180 at A1 and 181 at A2 of CON2.

9. Remove CON2 from unit and prepare to install the replacement parts. Replace old contactor with a new contactor. Re-connect the wires on CON1 and or CON2 based on schematic diagram. Tighten all connections but do not overtighten as equipment damage may occur.

4.1.5 Replacement of Fans (FAN1 & FAN2)

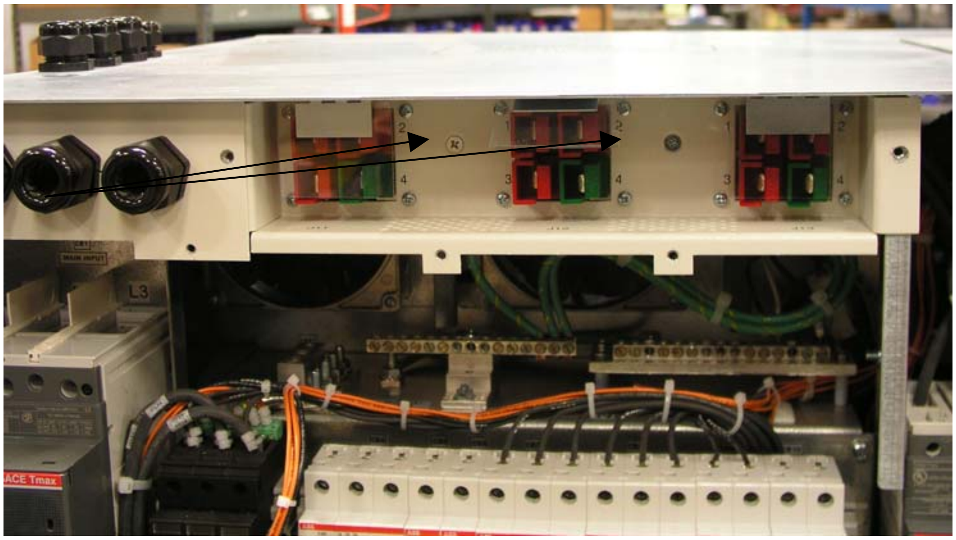
If the fans are deemed defective as per the Troubleshooting Guide in 4-1.8 follow the steps below to replace the fans. The fans are mounted behind the circuit breaker panel of the PDU located on top of the unit.

Fans or assemblies with static sensitive devices will be packaged in accordance to prevent damage due to ESD (Electrostatic Discharge). Procedures to prevent ESD should be followed when handling static sensitive devices.

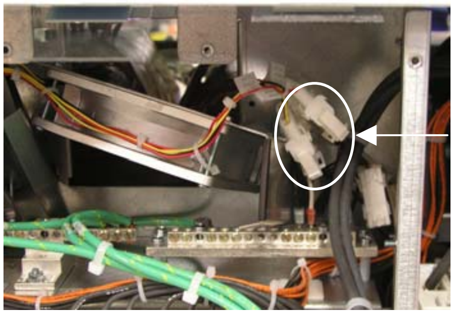
Tools needed: Phillips screwdriver or power screwdriver set.

1. Disconnect XPS power cables (J11, J12 and J13).
2. Remove 2 (10-32x1/2) screws at the Anderson assembly connector. Note: Do not discard screws, they will be used to replace assembly.

2 (10-32x1/2) screws

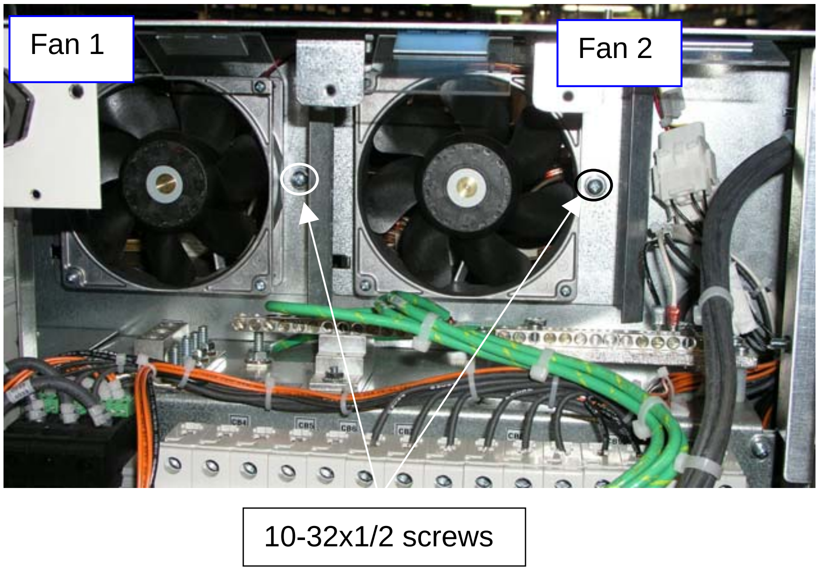


3. Unplug fan connectors at the quick disconnect module and remove fan assembly from unit. Cut tie wraps as needed.



Quick disconnect

4. Remove fans (FAN1 and FAN2) by unscrewing 2 (10-32x1/2) screws using Phillips screwdriver. Note: Do not discard screws, they will be used to replace fan assembly.



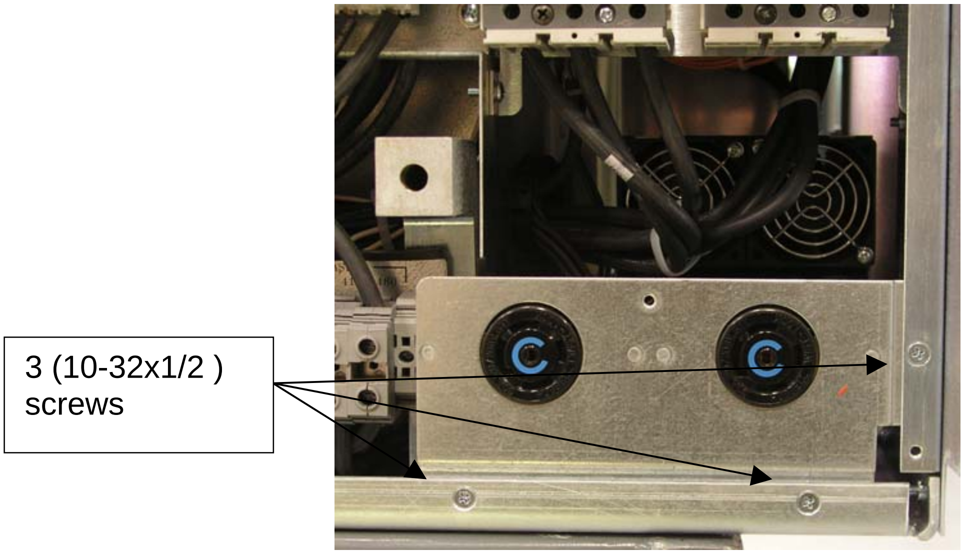
5. Replace old fan assembly with FRU using the screws previously removed. Re-connect housing connectors P14 with J17 and P19 with J19.

4.1.6 Replacement of Power Supplies (PS1 & PS3)

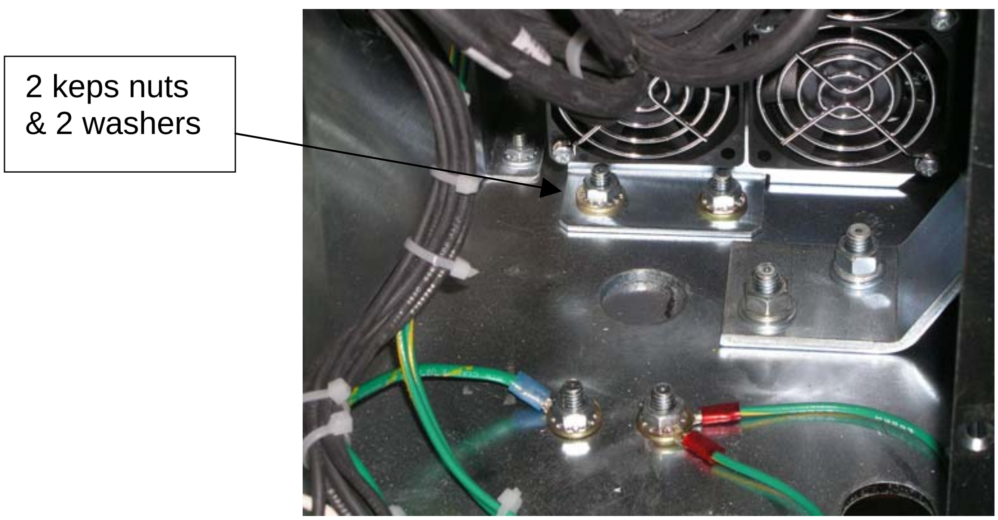
If the power supplies are deemed defective as per the Troubleshooting Guide in 4-1.8 follow the steps below to replace the power supplies. The power supplies are mounted behind the receptacle assembly of the PDU located on bottom of the unit.

Tools needed: Phillips screwdriver or power screwdriver set, tie wraps.

1. Remove the 3 (10-32x1/2) screws at the receptacle assembly. Move assembly out of the way.

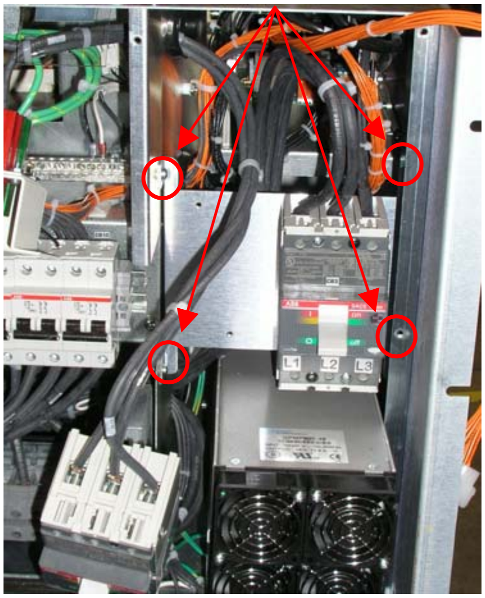


2. Remove 2 keps nuts and 2 washers at the power supply assembly.



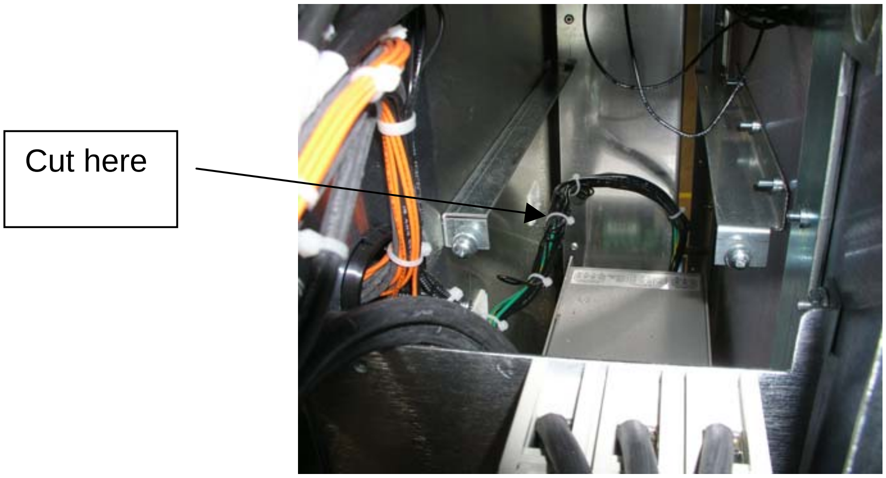
3. Disconnect CB1 and CB2 connections at top.

4. Remove breakers from bracket and move them out of the way.
5. Disconnect bracket (4 screws) and move it out of the way.

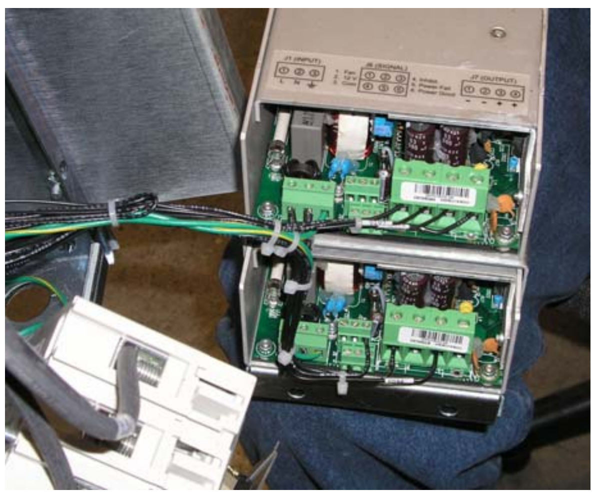


6. Slide power supplies up and out of PDU.

7. Use small cutter to (1) tie wrap at the power supply harnesses.

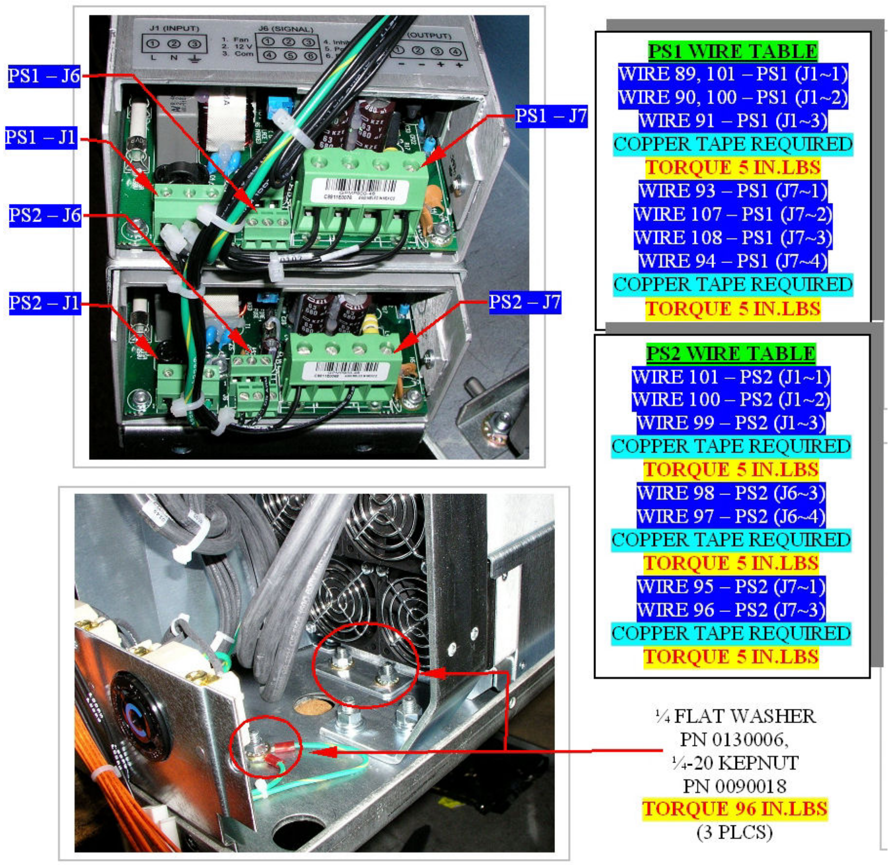


8. Disconnect ALL wires from the power supply assembly. Make sure you mark the wires with a masking tape and marker.



9. Reconnect wires to the new power supply assembly. Replace by power supply by performing these instructions in the reverse order.

10. Following are connection directions and recommended torque specifications for replacing the power supply.



4.1.7 Placement of New FRUs

1. To replace the FRUs in Sections 4.1.3 through 4.1.5 following the instructions in reverse order.

2. Use the schematic as needed to ensure placement of wires and/or cables.

3. To close the unit back up following the directions of Section 4.1.2 in reverse order.

4. Once unit has been closed up turn on Main Input circuit breaker.

4.1.8 Troubleshooting Guide

The following table lists fault conditions that may be encountered along with possible causes and solutions.

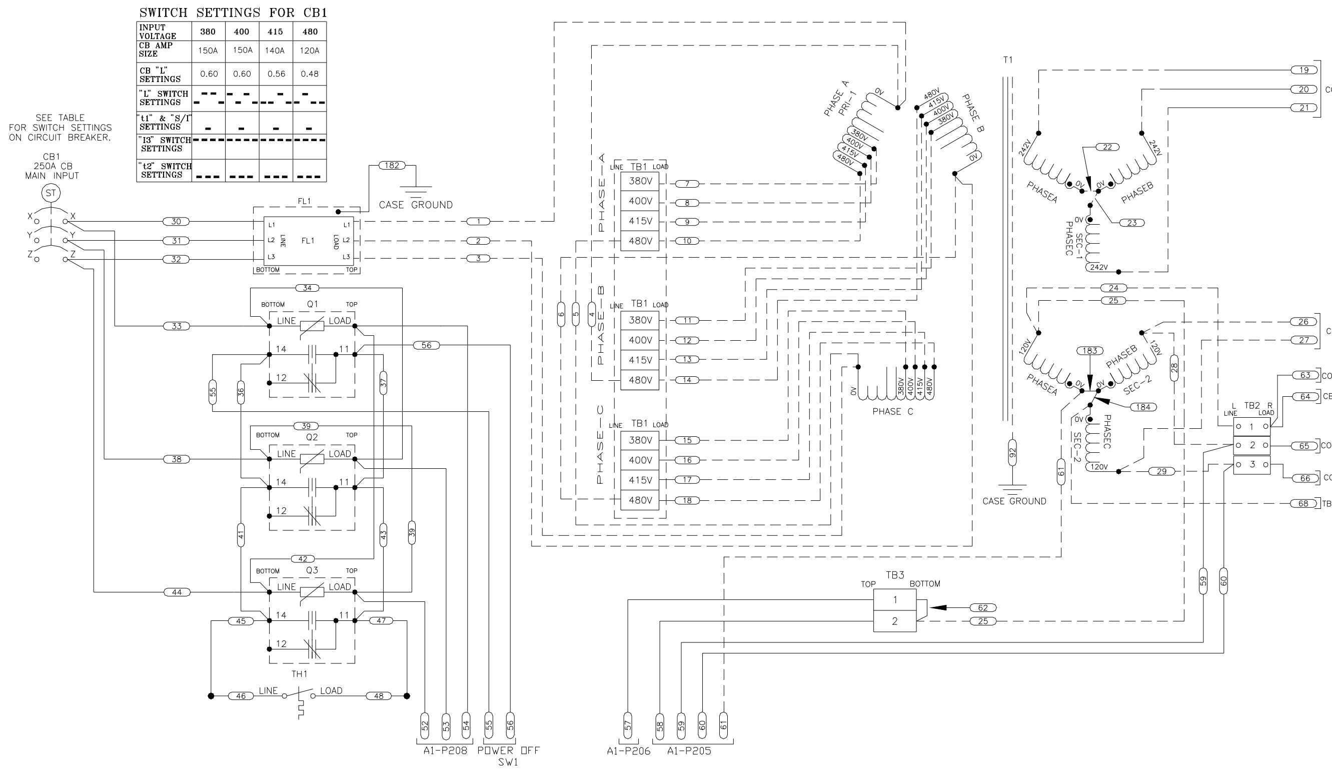
Troubleshoot Guide
Table 4-2

MODULE	SYMPTOM	CAUSE	SOLUTION
PDU	Main Input CB Trips	Overtemperature	1. Investigate cause of overtemperature 2. Allow unit to cool down. 3. Reset the circuit breaker
		Shorted Power Off Switch	1. Turn power off 2. Reset Main Circuit Breaker 3. Turn power on 4. If CB trips, call for support
		Open from J3-2 to J3-3	1. Disconnect cable at J3 2. With DVM check for open at cable connector P3-2 to P3-3.
	Distribution Breakers Tripping	Defective Circuit Breaker	1. Turn power off 2. Check if circuit breaker still trips (won't latch up)
		Overload	1. Check load by disconnecting cabinet from PDU whose circuit breaker is tripping, or use an Amp Clamp / DVM to check current draw. 2. Correct load problem.
	CON1/CON2 Contactor not operating	Defective Contactor	1. Ensure that the contactor CON1 is receiving 120V to the coils. 2. If 120V is present, replace the contactor. 3. If 120V is not present, verify that CON2 is receiving 120V to the coils 4. If 120V is present, replace the contactor. 5. If 120V is not present, replace the PGR cabinet.
		Defective Control Board A1	1. Disconnect cable at J2. 2. Check at cable connector if P2-7 to P2-8 is shorted. 3. If shorted, replace PDU 4. If not shorted, connect test plug and verify PDU operation. 5. If PDU operates with Test Plug, then replace PGR cabinet. 6. If PDU does not operate with Test Plug, then replace Control Board A1
	Indicators not lighting when power is applied	Input Circuit Breaker off	1. Turn Input Circuit Breaker, CB1, on.
		Input Phase Missing	1. Check that other phase LED's are on. 2. Measure at TP
		Light Burned Out	1. Check that other phase LED's are on. 2. Check condition of light
		Not Getting Proper Output	1. Measure input and output, at front panel test points, of each phase, from phase to phase (208 VAC: 208 VAC) and phase to ground (120 VAC: 120 VAC). 2. If difference is not a 100:1 ratio, replace PGR cabinet

Troubleshoot Guide
Table 4-2 (cont.)

MODULE	SYMPTOM	CAUSE	SOLUTION
PDU	Unit does not exit ESTOP mode (cannot turn on output contactor)	Board not receiving reset signal	1. Try resetting the unit by pressing SW1 on the front panel.
		Board not receiving power	1. Confirm connection on the board 2. Confirm board is receiving 120V by measuring at P6-11(hot) and P6-9 (neutral)
	Fans not running	Power supply not receiving power	1. Ensure CB9 is in the ON position 2. Measure between A1 and A2 at J4 on the PTC board. This should measure +48Vdc whenever the power supply is receiving power.
		Defective power supply	1. Check connection to the power supply. If power supply has good input connection the power supply is likely defective.
		Defective Fan	1. Check connection of fan. If power supply is deemed good ad fan has good connection fan is likely defective.

SCHEMATICS


SWITCH SETTINGS FOR CB1
INPUT VOLTAGE 380 400 415 480
CB AMP SIZE 150A 150A 140A 120A
CB "L" SETTINGS 0.60 0.60 0.56 0.48
"L" SWITCH SETTINGS
"t1" & "S/T" SETTINGS
"I3" SWITCH SETTINGS
"t2" SWITCH SETTINGS
SEE TABLE FOR SWITCH SETTINGS ON CIRCUIT BREAKER.
CB1 250A CB MAIN INPUT
FL1
CASE GROUND
Q1
Q2
Q3
TH1
TB1
PHASE A
PHASE B
PHASE C
PRI-1
T1
SEC-1
SEC-2
TB2
TB3
CON1
CON2
CB9
CB6
TB5
A1-P208
POWER OFF SW1
A1-P206
A1-P205

TEAL PDU MODULE IN XGD CABINET